GNSS Data Access Guide

Prepared for

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Oregon Department of Transportation

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Introduction

This guide contains instructions for accessing, and visualizing near real-time RTK-GNSS data from a PostgreSQL database hosted on the Microsoft Azure platform. Section 1 outlines the steps to download and install the required software (PostgreSQL Command Line Tools). Section 2 outlines the steps to setup PostgreSQL Command Line Tools in the Windows operating system. Section 3 covers how to use the provided PowerShell (.ps1) scripts to retrieve the near real-time data into a comma-separated values file (.csv). Section 4 covers how to import the .csv files generated in Section 3 into a provided Microsoft Excel template and calculate the displacement rates. Section 5 includes additional information about the GNSS data management processing pipeline, as well as the PostgreSQL database setup.

Included Assets

The following items were electronically delivered to Oregon State University (OSU) and the Oregon Department of Transportation (ODOT) as part of this project.

1. GNSS_Data_Access_Guide.pdf (this document)
2. get_gnss_data.ps1
    a. PowerShell file that can be used to retrieve GNSS data for a specific rover/landslide site combo into a csv file
3. get_azi_data.ps1
    a. PowerShell file that can be used to retrieve GNSS data for each rover at the Arizona Inn Landslide into a csv file
4. get_hoos_data.ps1
    a. PowerShell file that can be used to retrieve GNSS data for each rover at the Hooskanaden Landslide into a csv file
5. save_csv.sql
    a. SQL script used by the above PowerShell scripts to export the PostgreSQL data into a csv file.

6. GNSS_VIEWER.xlsx
    a. A Microsoft Excel File that can be used as a template to import GNSS csv data and compute displacement rates.
7. LICENSE
    a. File dictating the specific licensing covering each of the above materials.

1 Download & Install PostgreSQL

This section outlines the steps to download and install PostgreSQL Command Line Tools that are required to run the PowerShell scripts that download the near real-time GNSS data. This step only needs to be completed once on each workstation that will be used to run thePowerShell scripts, assuming PostgreSQL is not already installed. Note: you may need a member of your IT team to complete these steps on your behalf. These steps assume you are using the Windows operating system.

1. Download the PostgreSQL installer.
    a. https://www.postgresql.org/download/windows/
2. Select “Download the installer”.
3. Download the installer for version 14.8 or higher.
4. Once downloaded, run the installer and follow the onscreen instructions.
5. Under the “Select Components” section, only the “Command Line Tools” are required to be installed.
6. PostgreSQL Command Line Tools should now be installed.

2 Setting up PostgreSQL Command Line Tools

This section outlines the steps to setup PostgreSQL Command Line Tools that are required to run the PowerShell scripts that download the near real-time GNSS data. This step will make the PostgreSQL Command Line Tool (psql command) callable from PowerShell and the command prompt. This step only needs to be completed once on each workstation that will be used to run the PowerShell scripts. Note: you may need a member of your IT team to complete these steps on your behalf. These steps assume you are using the Windows operating system.

1. Add the path to PostgreSQL tools to the system environment variables.
    a. Open the command prompt as administrator, enter the following command:
        i. *setx /M path "%PATH%;C:\Program Files\PostgreSQL\{VERSION}\bin"*
        ii. Where VERSION is the version of PostgreSQL you installed in section 1.
        iii. E.g *setx /M path "%PATH%;C:\Program Files\PostgreSQL\15\bin"*
    b. In the Windows search bar search and open the “Edit the System Environment Variables” control panel settings.
    c. Click “Environment Variables…” in the bottom right.
    d. Select the “Path” Variable under System Variables, and click Edit.
    e. Click new, then copy the path to the postgreSQL bin directory.
        i. C:\Program Files\PostgreSQL\{VERSION}\bin
    f. Click ok to exit the environment variables/system properties.
    g. Open a new command prompt or powershell instance and type the follow command:
        i. *psql --version*
    h. This should verify that psql is available.

3 Retrieving Data from the PostgreSQL Databases

This section covers how to use the provided PowerShell (.ps1) scripts to download the current up to data near real-time GNSS data into comma-separated values files (.csv). These steps should be completed each time a user wishes to download a new instance of the database tables. Following the instructions in this section will generate a csv file for each rover for each landslide site.

3.1 Retrieving all Rover Data from Arizona Inn

1. Browse to the directory containing the PowerShell scripts (*.ps1) and SQL script (save_csv.sql) delivered along with this guide.
2. Right click on the “get_azi_data.ps1” file, and select “Run with PowerShell”.
3. A PowerShell window should appear and incrementally download data for each rover into its own csv file.
4. That's it! There should now be 4 new csv files in the directory with the following names:
    a. YYYYMMDD_azi_r1.csv
        i. csv data containing GNSS data for r1
    b. YYYYMMDD_azi_r2.csv
        i. csv data containing GNSS data for r2
    c. YYYYMMDD_azi_r3.csv
        i. csv data containing GNSS data for r3
    d. YYYYMMDD_azi_r4.csv
        i. csv data containing GNSS data for r4
5. Where YYYYMMDD is the date that the script was run.
6. Each of these files will contain the full time-series of the data from installation data to the time that the “get_azi_data.ps1” script was executed.

3.2 Retrieving all Rover Data from Hooskanden

1. Browse to the directory containing the PowerShell scripts (*.ps1) and SQL script (save_csv.sql) delivered along with this guide.
2. Right click on the “get_hoos_data.ps1” file, and select “Run with PowerShell”.
3. A PowerShell window should appear and incrementally download data for each rover into its own csv file.
4. That's it! There should now be 6 new csv files in the directory with the following names:
    a. YYYYMMDD_hoos_r1.csv
        i. csv data containing GNSS data for r1
    b. YYYYMMDD_hoos_r2.csv
        i. csv data containing GNSS data for r2
    c. YYYYMMDD_hoos_r3.csv
        i. csv data containing GNSS data for r3
    d. YYYYMMDD_hoos_r4.csv
        i. csv data containing GNSS data for r4
    e. YYYYMMDD_hoos_r5.csv
        i. csv data containing GNSS data for r5
    f. YYYYMMDD_hoos_r6.csv
        i. csv data containing GNSS data for r6
5. Where YYYYMMDD is the date that the script was run.
6. Each of these files will contain the full time-series of the data from installation data to the time that the “get_hoos_data.ps1” script was executed.

3.3 BONUS: Specifying the time range

If you would like to select a specific time range, rather than downloading all of the data, follow the following steps:

1. Browse to the directory containing the PowerShell scripts (*.ps1) and SQL script (save_csv.sql) delivered along with this guide.

2. In the File Explorer address bar type “powershell” and hit Enter. This should launch a PowerShell instance in the GNSS_data_access directory.
3. Drag and drop the “get_azi_data.ps1” file into the PowerShell window. This should enter the full path to the “get_azi_data.ps1” file.
4. Use the -time0 flag to indicate the start date you would like to download data for.
5. Use the -time1 to indicate the end data you would like to download the data for.
6. The full command should appear as below:
7. *C:\PATH\TO\FILES\get_azi_data.ps1 -time0 “‘YYYY-MM-DD’” -time1 “‘YYYY-MM-DD’”*
    a. Note the timestamps should be encompassed by both “” and ‘’ -> “ ‘ YYYYY-MM-DD ‘ “ (spaces added for easy visualization).
8. For example to retrieve all of the data for June of 2023, the following command could be used:
9. *C:\PATH\TO\FILES\get_azi_data.ps1 -time0 “‘2023-06-01’” -time1 “‘2023-07-01’”*
10. Note you can also run the same command with only the -time0 flag, or only the -time1 flag to only specify the start or end date.

3.4 OPTIONAL: Running the get_gnss_data.ps1

The above “get_azi_data.ps1” and “get_hoos_data.ps1” scripts call the “get_gnss_data.ps1” script for each rover at their respective sites. Optionally you can call the “get_gnss_data.ps1” from PowerShell directly as shown below:

1. *C:\PATH\TO\FILES\get_gnss_data.ps1 -site {SITENAME} -rover {ROVERNAME} -time0 “‘YYYY-MM-DD’” -time1 “‘YYYY-MM-DD’” -timezone {TIMEZONE}*
2. Below is a description of each of the arguments:
    a. -site

   i. Specifies the landslide site to fetch data from. This expects that a PostgreSQL database with this name exists.
   ii. E.g “azi”
b. -rover
   i. Specifies the rover to fetch data from. This expects that a PostgreSQL table with this name exists in the site database.
   ii. E.g “r1”
c. -time0
   i. The start date you would like to download data after.
   ii. E.g “‘2023-01-01’”
d. -time1
   i. The end date you would like to download data before.
   ii. E.g “‘2023-12-31’”
e. -timezone
   i. The timezone you would like the timestamps to be in
   ii. E.g “‘US/Pacific’”

3.5 Further Reading

The above section describes how to query and download data to a csv in near real-time from the database using the provided tools. For further reading on developing advanced data queries and operations, the user can reference the PostgreSQL documentation. Further information about the specific PostgreSQL instance and setup can be found in Tables 5.2 and 5.3.

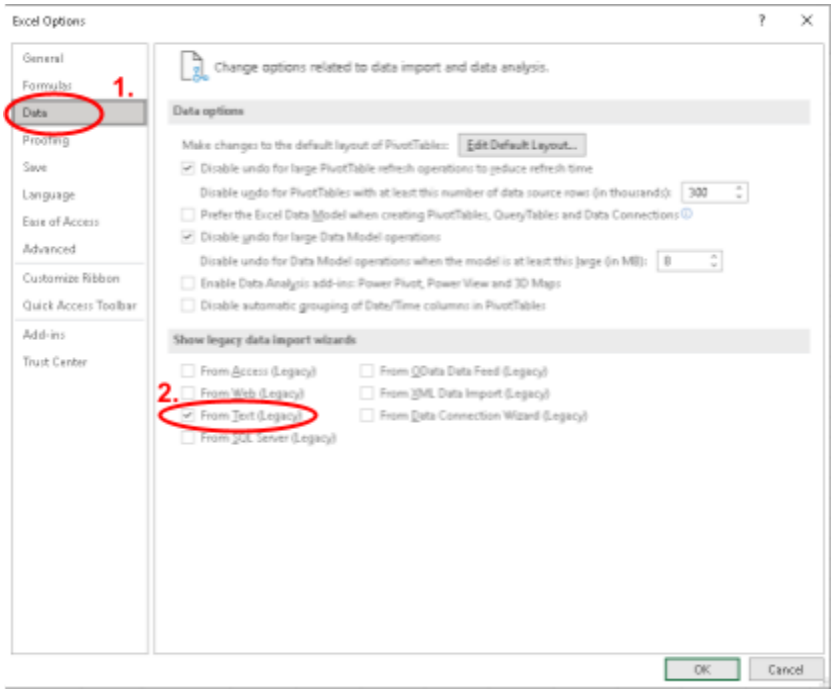
4 Importing Data into the Microsoft Excel Template

Once data has been downloaded from the PostgreSQL database as outlined in Section 3, the data can be imported into the provided Microsoft Excel Template to compute the displacement rates, and perform further data analysis. This section contains a guide to importing the csv data into the provided Microsoft Excel template, as well as an overview of the template layout.

4.1 Importing the csv Data into the Excel Template

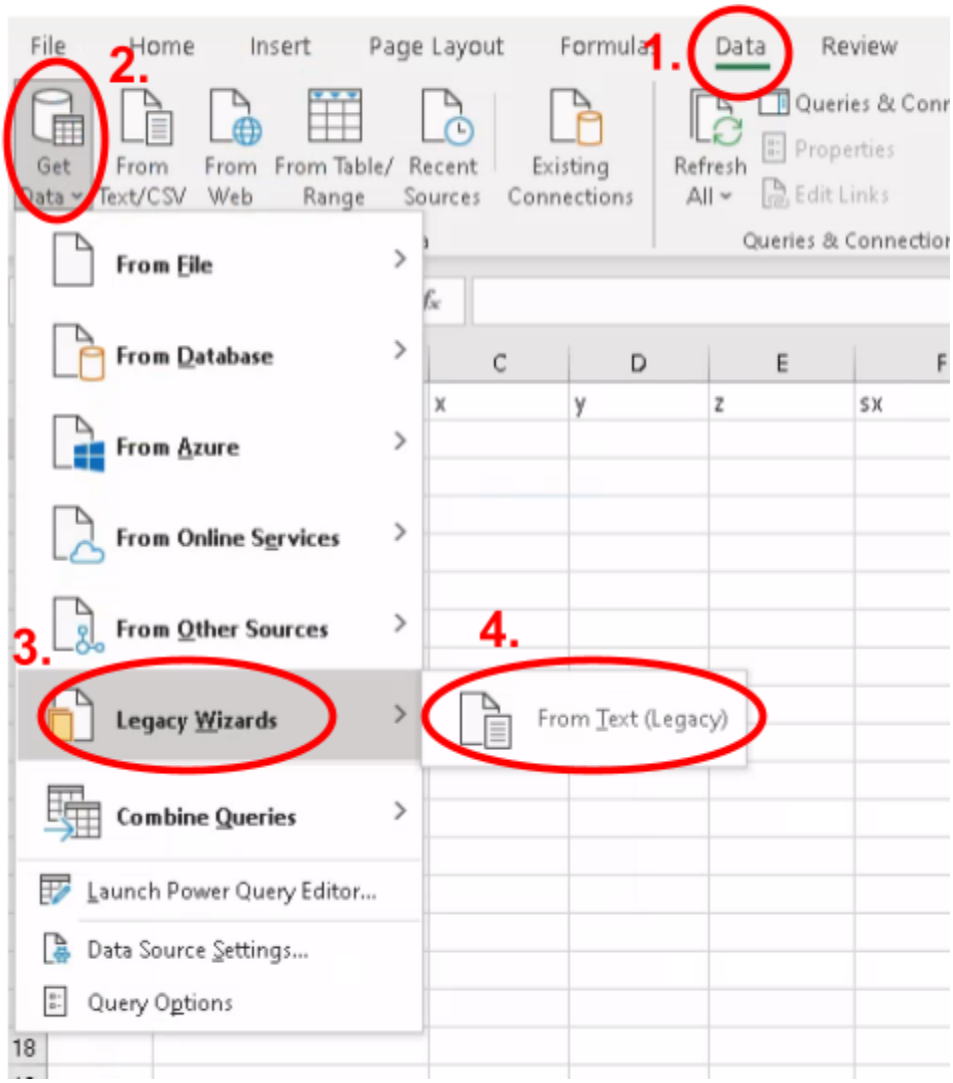
To import csv data into the Microsoft Excel Template follow the steps below:

1. Open the “GNSS_VIEWER.xslx” Excel spreadsheet template.
2. At the bottom, click on the “COORDINATES” sheet.
3. This sheet is where you will import the csv file, not the column headings in this sheet correspond to the PostgreSQL table structure outlined in Section 5.3.
4. If not already, enable the legacy “From Text” wizard as described below:

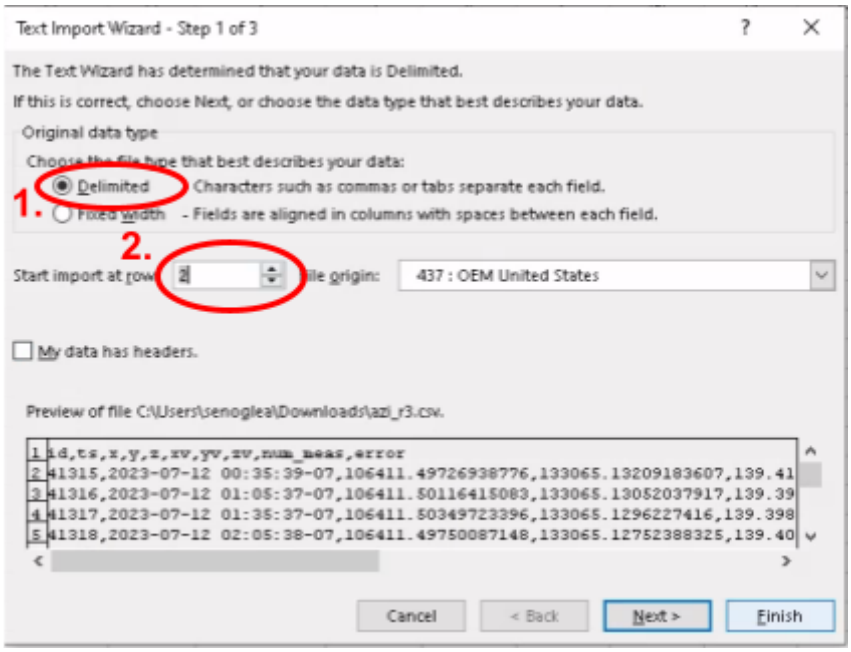


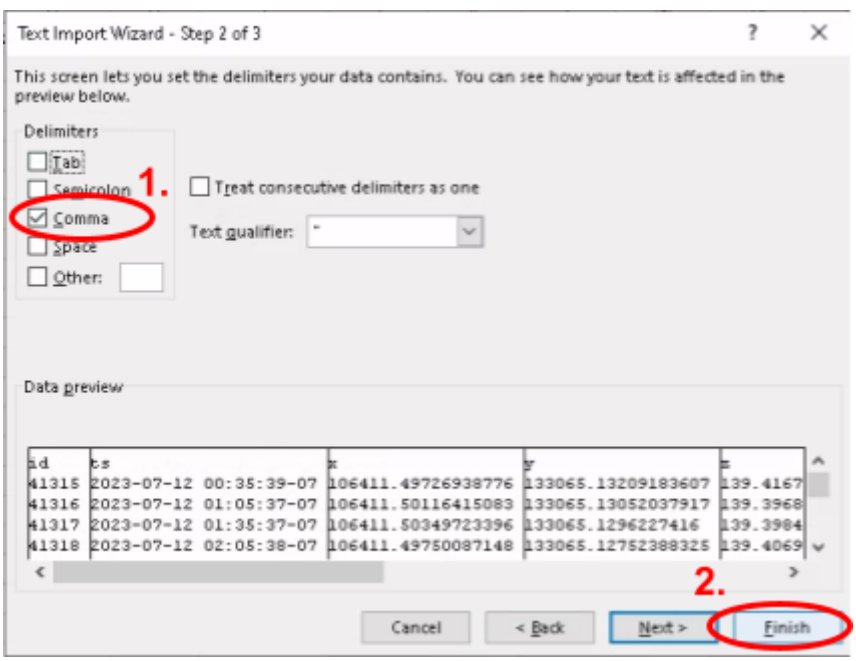
a. In the Navigation bar, click “File”, then click “options” on the bottom left corner.
b. In the pop-up, click the “Data” tab on the left.
c. Under “Show legacy data import wizards”, check the box next to “From Text (Legacy)”, then click “Ok” to close the window.

5. To import the data, select the empty cell in the top left of this sheet (A2), then in the top navigation bar, under data, in the “Get & Transform Data” section, click the “Get Data” dropdown, then in “Legacy Wizards” select “From Text (Legacy)” as shown below.



6. In the pop up window, navigate to the csv file, and click open.
7. In step 1, select “Delimited”, and “Start import at row 2”, then click next.
8. In step 2, check “Comma” under “Delimiters” then click “Finish”.
9. In the new pop up, ensure “Existing worksheet” is selected under “Where do you want to put the data?”, and make sure cell A2 is selected. Click ok.





10. The coordinate values and associated metadata contained in the csv will now appear in the sheet, and be available for further analysis.

4.2 Viewing the Displacement Computations

Once the csv data has been imported into the Microsoft Excel Template as described above, we can view the displacement computations following the steps below:

1. At the bottom, click on the "DISPLACEMENTS" sheet.
2. This will open a sheet containing a preformatted table with the displacement rates based on the csv data imported in Section 4.1.
3. The column values in the table are described below:
    a. **ID**: The data entry number representing the order data was added to the database.
    b. **TS**: The timestamp of the data in MM/DD/YYYY HH:MM form
    c. **ΔX**: The cumulative displacement between the current and first reading along the X-axis. Negative indicates west movement.
    d. **ΔY**: The cumulative displacement between the current and first reading along the Y-axis. Negative indicates south movement.
    e. **ΔZ**: The cumulative displacement between the current and first reading along the Z-axis. Negative indicates down movement.
    f. **Δ2D**: The magnitude of cumulative displacement between the current and first reading along the horizontal plane.
    g. **Δ3D**: The magnitude of cumulative displacement between the current and first reading in 3D space.
    h. **σX**: The standard deviation of the X coordinate.
    i. **σY**: The standard deviation of the Y coordinate.
    j. **σZ**: The standard deviation of the Z coordinate.
    k. **# Meas**: The number of measurements made during reading.
    l. **Error**: The errors encounter (if any). See Table 5.3 for details.
4. This displacements table can be used to quickly analyze the displacement experienced at a given rover between the specified dates.
5. The displacement data can also be used to generate cumulative displacement plots, or feed into further analysis to compute things like velocity, etc.

4.3 Changing Settings

4.3.1 Changing the Displacement Units

To change the measurement units that the displacement data are displayed in, follow the steps below:

1. At the bottom, click on the “SETTINGS” sheet.
2. In cell B6 enter “ft” if you would like to use feet as the measurement unit, or enter “m” if you would like to use meters as the measurement unit.
3. This should update the measurement of the displacement data displayed in the table in the “DISPLACEMENTS” sheet.

4.3.2 Changing the Error Code Filter

The error code filter can be used to ignore all readings with an error code value above a threshold. To change, follow the steps below:

1. At the bottom, click on the “SETTINGS” sheet.
2. In cell B7 enter the upper bound of measurement error codes that you would like to keep in your analysis.
3. This will automatically convert readings with a higher error code to NaN values in the “DISPLACEMENTS” sheet.
4. See Table 5.3 for a description of the meaning of each of the error values. A value of 0 or 1 is recommended as the default.

4.4 Plotting Displacement Data

This section covers creating a simple cumulative displacement plot of the 3D displacement contained in the “DISPLACEMENTS” sheet.

1. If not already, follow the steps above to import the csv data for the rover, landslide site, and timeframe that you would like to visualize.
2. At the bottom, click on the “DISPLACEMENTS” sheet.
3. Highlight the TS column for the timeframe you would like to create the plot for, by clicking the first cell, and dragging to the last cell.

4. Now highlight the Δ3D column by holding down the “ctrl” key, and selecting the corresponding cells in the Δ3D column.
5. In the top navigation bar, under insert, click on the desired chart you would like to make, in this example, we will make a line chart.
6. This will create a new chart window displaying a plot of the cumulative displacement across the time interval selected as shown in Figure 4.4 below.
7. You can now perform additional formatting, labeling, and editing to the plot to improve its readability as you see fit.
8. Note you can make the same plots for cumulative displacement across different coordinate axes, by selecting the corresponding columns instead of Δ3D.

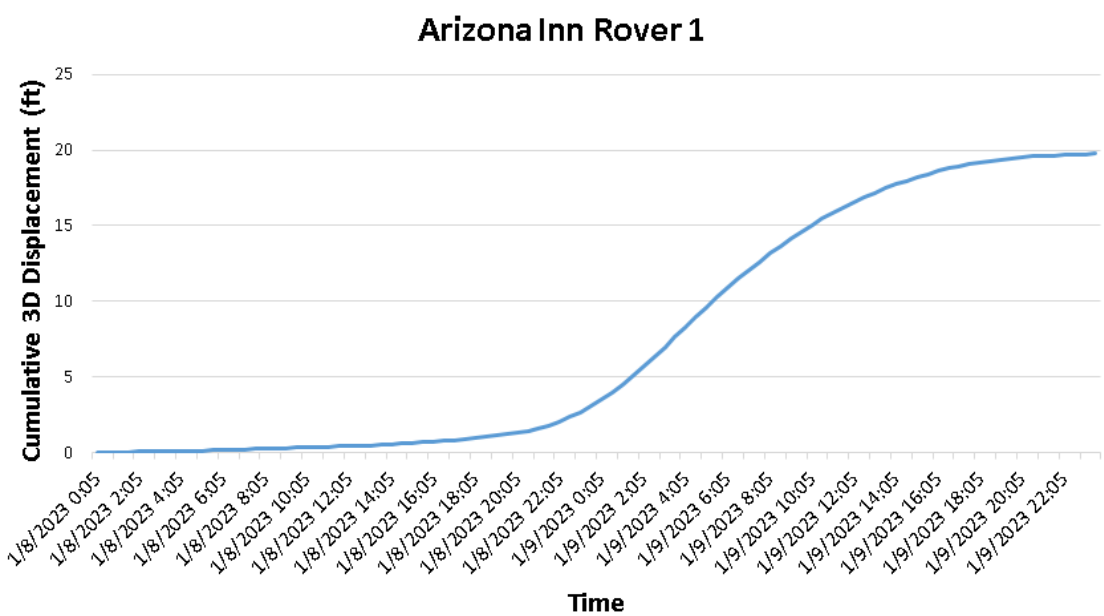


Figure 4.3: Example line chart showing cumulative 3D Displacement of Arizona Inn Rover 1 Created using the Microsoft Excel GNSS_VIEWER.xlsx Template.

4.5 Further Analysis

In addition to the visualization and analysis shown above, further analysis can be developed using Microsoft Excel, or other tools capable of handling csv data. This further analysis could include, but is not limited to: velocity computation, acceleration computation, data smoothing using windowed functions, forecasting etc.

5 Additional Information

The following section contains additional documentation about how the deployed GNSS data management system operates as well as information about the specific PostgreSQL instance that may be useful for creating further query/analysis. Note this can all be interactively explored through the PostgreSQL Command Line Tool.

5.1 Theory of Operation

The below section contains a high level overview of the theory of operation outlining the data pipelines from raw data being uploaded by the base, to GNSS data being downloaded by the user.

To setup the GNSS data management and processing, an Azure resource group with the following resources was created:

1. Azure IoT Hub
2. Azure Functions
3. Azure Database for PostgreSQL (Flexible Server)

The GNSS Data is sent via a webhook from the Base station to the Azure IoT Hub. This triggers an Azure Event Hub Trigger that triggers the data_input function in the Azure Functions resource to collect the webhook data and process the raw sensor data provided by the base into a projected coordinate system (OCRS Oregon Coast meters). These data are then posted to the table belonging to the corresponding rover within the database belonging to the landslide site. The provided PowerShell scripts can then be executed by a user to download the data for each rover into a csv file, which can be brought into the provided Microsoft Excel Template to compute the displacement rates, and conduct further analysis. This overview is shown as a flowchart in Figure 5.1 below:

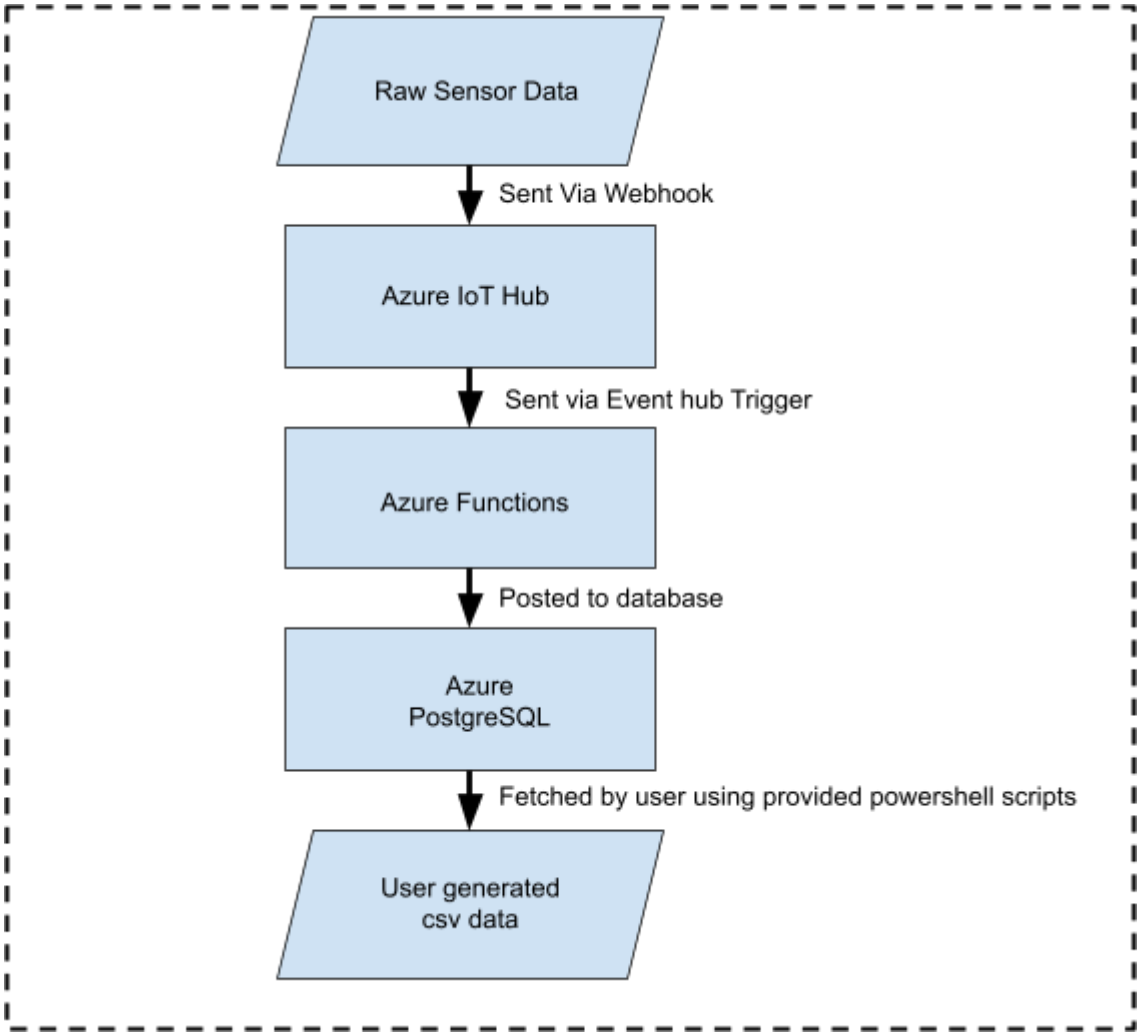


Figure 5.1: Flowchart showing the high level overview of the data processing pipeline from data upload by a base station to data download by a user.

5.2 Existing PostgreSQL databases

Table 5.2: The existing PostgreSQL databases and their parameters.

Landslide Site	Host	Port	Database Name	Tables
Arizona Inn	db-engr-cce-gnss-db1.postgres.database.azure.com	5432	azi	r1, r2, r3, r4
Hooskanaden	db-engr-cce-gnss-db1.postgres.database.azure.com	5432	hoos	r1, r2, r3, r4, r5, r6

5.3 PostgreSQL Table Structure

Table 5.3: The PostgreSQL data types and a description of the columns that make up each table in each database.

Column	PostgreSQL Data Type	Description
id	serial	Auto incrementing integer representing the order data was added to the database.
ts	timestamptz	ISO string representing the data timestamp with time zone.
x	double	X coordinate of the rover position.
y	double	Z coordinate of the rover position.
z	double	Y coordinate of the rover position.
sx	double	Standard deviation of the X coordinate.
sy	double	Standard deviation of the Y coordinate.
sz	double	Standard deviation of the Z coordinate.
num_meas	small int	Number of measurements made during reading.
error	small int	Error encounter (if any): 0 = no error, 1 = max number of readings was not achieved, 2 = floating fix allowed in solution, 3 = no RTK fix was achieved, 4 = no time fix was achieved, 5 = no GNSS fix was achieved, 6 = no communication received.

5.4 PostgreSQL Credentials

Users have been provided with a set of credentials (username and password) to access the above PostgreSQL databases based on their institution (OSU or ODOT). These credentials are stored in the “get_gnss_data.ps1” script, but can additionally be used for accessing the databases using the PostgreSQL Command Line Tool, or any other software capable of accessing a PostgreSQL server database. Anyone that gets access to these credentials will be able to read data from the databases. Note: the credentials provided are read-only, meaning they can read data from the database, but cannot write data to the database. This has been purposefully set to prevent any accidental writes or deletes that may corrupt the data. Only the Azure Functions that collect and write the incoming raw data to the databases has access to write privileges.